The Quant Interview Survival Guide:

108 Elite-Level Problems for Tier-1 Firms

Goldman Sachs · JP Morgan · Morgan Stanley · Bank of America
Barclays · UBS · Jane Street · Citadel

Formula-Driven Solutions with Memorization Frameworks

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Goldman Sachs's Conditional Variance

Show that $Var(X) = E[Var(X|Y)] + Var(E[X|Y])$. Apply this to X = portfolio return, Y = market regime (Bull/Bear with probabilities 0.7/0.3). In Bull, $X \sim N(10\%, 15\%)$. In Bear, $X \sim N(-5\%, 25\%)$. Compute total variance.

Law of total variance:

$$\text{Total Var} = E[\text{Conditional Var}] + \text{Var(Conditional Mean)}$$

Compute:

$$E[X|Y = \text{Bull}] = 10\%$$
$$E[X|Y = \text{Bear}] = -5\%$$

$$E[X] = 0.7 \times 10\% + 0.3 \times (-5\%) = 7\% - 1.5\% = 5.5\%$$

$$Var(E[X|Y]) = E[(E[X|Y] - E[X])^2]$$

$$= 0.7(10\% - 5.5\%)^2 + 0.3(-5\% - 5.5\%)^2$$

$$= 0.7(4.5\%)^2 + 0.3(-10.5\%)^2 = 0.7 \times 0.002025 + 0.3 \times 0.011025$$

$$= 0.0014175 + 0.0033075 = 0.004725 = 0.4725\%^2$$

Conditional variance:

$$E[Var(X|Y)] = 0.7 \times 15\%^2 + 0.3 \times 25\%^2$$

$$= 0.7 \times 0.0225 + 0.3 \times 0.0625 = 0.01575 + 0.01875 = 0.0345 = 3.45\%^2$$

Total variance:

$$Var(X) = 3.45\%^2 + 0.4725\%^2 = 3.9225\%^2$$

Volatility:

$$\sigma = \sqrt{0.039225} = 19.8\%$$

Total Variance Decomposition

$$\sigma^2_{\text{total}} = \underbrace{E[\sigma^2_{\text{regime}}]}_{\text{Average Risk}} + \underbrace{\sigma^2_{\text{mean return}}}_{\text{Regime Uncertainty}}$$

Memory Anchor: Total variance scales **additively** with average conditional variance and variance of conditional means. Regime switching adds risk even if regime probabilities are known.

Morgan Stanley's Simpson's Paradox

A hedge fund has two strategies A and B. Last year: Strategy A had 1000 trades with 60% win rate. Strategy B had 100 trades with 70% win rate. This year: Strategy A had 100 trades with 50% win rate. Strategy B had 1000 trades with 60% win rate. Show that combined win rate is higher for each strategy individually but lower overall for the fund. Explain why.

Last year: - A wins: 600/1000 = 60- B wins: 70/100 = 70- Combined: 670/1100 = 60.9

This year: - A wins: 50/100 = 50- B wins: 600/1000 = 60- Combined: 650/1100 = 59.1

Now both strategies individually performed worse this year (A: 60

Actually need to construct true paradox:

Last year: - A: 60/100 = 60- B: 7/10 = 70- Combined: 67/110 = 60.9

This year: - A: 50/100 = 50- B: 540/900 = 60- Combined: 590/1000 = 59

Both strategies declined but overall increased? Let's fix:

Better example: Last year: - A: 600/1000 = 60- B: 8/10 = 80- Combined: 608/1010 = 60.2

This year: - A: 50/100 = 50- B: 720/1000 = 72- Combined: 770/1100 = 70.0

Now A declined (60

Inclusion-Exclusion

$$P(\cup E_i) = \sum P(E_i) - \sum P(E_i E_j) + \sum P(E_i E_j E_k) - \dots$$

Memory Anchor: Alternating sum **converges** to true probability. For independent events with small p, first-order approximation $1-(1-p)^n$ is excellent. Higher-order terms provide **exponentially** smaller corrections.

Citadel's Bayesian Updating with Beta

You model a strategy's win rate as $\theta \sim \text{Beta}(2,3)$. After 10 trades, you observe 7 wins, 3 losses. What's the posterior? What's the predictive probability that next trade wins? Show how this is a weighted average.

Prior: $\theta \sim \text{Beta}(\alpha = 2, \beta = 3)$ Likelihood: $\text{Binomial}(10, \theta)$ with 7 wins Posterior: $\theta|\text{data} \sim \text{Beta}(\alpha+7, \beta+3) = \text{Beta}(9,6)$

Posterior mean:

$$E[\theta|\text{data}] = \frac{\alpha+7}{\alpha+\beta+10} = \frac{9}{15} = 0.6$$

Predictive probability next trade wins:

$$P(\text{win next}|\text{data}) = \int_0^1 \theta \cdot f(\theta|\text{data}) d\theta = E[\theta|\text{data}] = 0.6$$

This is weighted average:

$$\frac{\alpha+\beta}{\alpha+\beta+n} \cdot \frac{\alpha}{\alpha+\beta} + \frac{n}{\alpha+\beta+n} \cdot \frac{k}{n}$$

$$= \frac{5}{15} \cdot \frac{2}{5} + \frac{10}{15} \cdot \frac{7}{10} = \frac{2}{15} + \frac{7}{15} = \frac{9}{15} = 0.6$$

Beta-Binomial Conjugate

$$\text{Posterior mean} = w \cdot \text{prior mean} + (1-w) \cdot \text{MLE}$$

Memory Anchor: Weight $w = \frac{\alpha+\beta}{\alpha+\beta+n}$ is **inversely proportional** to sample size. Small samples $\Rightarrow$ shrink to prior. Large samples $\Rightarrow$ converge to MLE.

UBS's Probability of Dominance

Stock A returns $r_A \sim N(10\%, 20\%)$. Stock B returns $r_B \sim N(12\%, 25\%)$ with correlation $\rho = 0.6$. What's $P(r_A > r_B)$? What's the probability A beats B by more than 5%?

Let $D = r_A - r_B$.

$$E[D] = 10\% - 12\% = -2\%$$

$$Var(D) = Var(r_A) + Var(r_B) - 2Cov(r_A, r_B)$$

$$= 0.20^2 + 0.25^2 - 2 \times 0.6 \times 0.20 \times 0.25$$

$$= 0.04 + 0.0625 - 0.06 = 0.0425$$

$$\sigma_D = \sqrt{0.0425} = 0.206 = 20.6\%$$

$$P(r_A > r_B) = P(D > 0) = P\left(\frac{D-(-2\%)}{20.6\%} > \frac{0-(-2\%)}{20.6\%}\right)$$

$$= P\left(Z > \frac{2}{20.6}\right) = P(Z > 0.097) = 0.461$$

Probability A beats B by >5

$$P(D > 5\%) = P\left(Z > \frac{5-(-2)}{20.6}\right) = P(Z > 7/20.6) = P(Z > 0.34) = 0.367$$

Connectivity Threshold

$$p_{\text{connect}} \propto \frac{\ln n}{n}$$

Memory Anchor: Threshold scales **logarithmically** with n. Each vertex needs about $\ln n$ neighbors to connect graph. Sparse graphs ($p < 1/n$) are disconnected; dense graphs ($p > \ln n/n$) are connected.

Jane Street's Stopping Time Problem

You flip a fair coin repeatedly. Let T be the first time you see the pattern HTH. What's $E[T]$? What's $E[T]$ for pattern HHH? Explain why they differ.

This is a pattern matching problem using Markov chains.

For pattern HTH, states represent matched suffix: - S0: no match - S1: H - S2: HT - S3: HTH (absorbing)

Transition matrix with expected time equations:

$$E_0 = 1 + 0.5E_1 + 0.5E_0$$

$$E_1 = 1 + 0.5E_1 + 0.5E_2$$

$$E_2 = 1 + 0.5E_3 + 0.5E_0$$

$$E_3 = 0$$

Solving: From $E_0 = 1+0.5E_1+0.5E_0 \Rightarrow E_0 = 2+E_1$ From $E_1 = 1+0.5E_1+0.5E_2 \Rightarrow 0.5E_1 = 1+0.5E_2 \Rightarrow E_1 = 2+E_2$ From $E_2 = 1 + 0.5(0) + 0.5E_0 = 1 + 0.5E_0$

Substitute: $E_1 = 2 + (1 + 0.5E_0) = 3 + 0.5E_0$ $E_0 = 2 + (3 + 0.5E_0) \Rightarrow 0.5E_0 = 5 \Rightarrow E_0 = 10$

For HHH: States: - S0: no match - S1: H - S2: HH - S3: HHH

$$E_0 = 1 + 0.5E_1 + 0.5E_0 \Rightarrow E_0 = 2 + E_1$$

$$E_1 = 1 + 0.5E_2 + 0.5E_0$$

$$E_2 = 1 + 0.5E_3 + 0.5E_0 = 1 + 0.5(0) + 0.5E_0$$

$$E_3 = 0$$

From $E_2 = 1 + 0.5E_0$ From $E_1 = 1 + 0.5(1 + 0.5E_0) + 0.5E_0 = 1 + 0.5 + 0.25E_0 + 0.5E_0 = 1.5 + 0.75E_0$ From $E_0 = 2 + 1.5 + 0.75E_0 \Rightarrow 0.25E_0 = 3.5 \Rightarrow E_0 = 14$

HTH takes 10 flips on average, HHH takes 14 flips. HTH has better overlap properties - when you fail to match, you retain partial progress.

Pattern Expected Waiting Time

$$E[T] = \frac{2^k}{\text{autocorrelation factor}}$$

Memory Anchor: Expected waiting time scales **exponentially** with pattern length but **inversely** with pattern overlap. HTH (non-overlapping) waits less than HHH (self-overlapping).

Citadel's Order Book Depth

The limit order book has bid prices $P_k = P_0 - k\delta$ for $k = 0, 1, ..., N$ with $\delta = \$0.01$. Each price level has independent depth $D_k \sim \text{Poi}(\lambda_k)$ with $\lambda_k = \lambda_0 e^{-k\alpha}$. What's the expected total quantity available below price $P_0 - K\delta$? What's the probability the best bid is exactly \$100.00 given $P_0 = 100.01$?

Expected total quantity below price $P_0 - K\delta$:

$$E[Q_K] = \sum_{k=K+1}^{N} E[D_k] = \sum_{k=K+1}^{N} \lambda_0 e^{-k\alpha}$$

$$= \lambda_0 e^{-\alpha(K+1)} \frac{1 - e^{-\alpha(N-K)}}{1 - e^{-\alpha}}$$

As $N \to \infty$:

$$E[Q_K] = \lambda_0 \frac{e^{-\alpha(K+1)}}{1 - e^{-\alpha}}$$

Probability best bid is exactly \$100.00: Best bid is highest price with non-zero depth.